



BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 1

Introduction to Function

Table of Contents

SL No	Topic	Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction	-	-	4-5
1.1	Learning Objectives	-	-	
2	Relation, Pictorial Diagrams, Domain, Co-Domain, Range of a Relation	-	-	6-10
2.1	Relations	-	-	
2.1.1	Types of Relations	-	-	
2.2	Pictorial Diagrams	-	-	
2.3	Domain	-	-	
2.4	Co-Domain	-	-	
2.5	Range of a Relation	-	-	
3	Function	-	-	11-12
4	Domain and Range of a Function	-	-	13-20
4.1	Finding Domain and Range of a Linear Function	-	-	
4.2	Types of Functions	-	-	
4.2.1	Real-Valued Functions	-	-	
4.2.2	Constant Function	-	-	
4.2.3	Identity Function	-	-	
4.2.4	Polynomial Function	-	-	
4.2.5	Rational Function	-	-	
4.2.6	Modulus Function (Absolute Value)	-	-	
4.2.7	Signum Function (Sign Function)	-	-	
4.2.8	Exponential Function	-	-	
4.2.9	Logarithmic Function	-	-	

	4.2.10	Greatest Integer Function (Step Function)	-	-	
5		Properties of Functions	-	-	21-23
6		Summary	-	-	23-24
7		Self-Assessment Questions	-	1	24-27
8		Terminal Questions	-	-	27-28
9		Answers	-	-	28-30



1. INTRODUCTION

The profound importance of mastering the fundamentals of Mathematics for students pursuing a Bachelor in Computer Applications (BCA) cannot be overstated. Mathematics forms the backbone of logical reasoning and analytical thought processes, which are indispensable in the field of computing and information technology. It equips students with the essential tools to tackle complex problems, devise algorithms, and understand the theoretical underpinnings of computer science. Particularly, the study of relations, sets, functions, and their properties lay down the groundwork for understanding more advanced topics such as database structures, programming, and software development. A strong foundation in these mathematical concepts not only enhances computational skills but also fosters a rigorous analytical mindset, enabling students to approach problems methodically and innovate solutions effectively.

Unit 1, which delves into the intricacies of relations, pictorial diagrams, domains, co-domains, and the range of a relation, stands as a critical component of the BCA curriculum. The concept of relations, at its core, establishes a framework to model and analyse the connections between different sets of data, an aspect central to computing and programming. By understanding how elements from one set relate to elements in another through various types of relations such as reflexive, symmetric, and transitive, students can better grasp data organization, database relationships, and algorithm efficiency. The exploration of pictorial diagrams further aids in visualizing these relationships, making abstract concepts more tangible and easier to comprehend. This unit not only lays the foundation for database management systems and data structures but also sharpens problem-solving skills by illustrating the underlying mathematics in computational processes. Therefore, a thorough understanding of this unit is paramount for any aspiring computer application professional, as it equips them with the necessary tools to model, analyse, and manipulate data effectively in various computing contexts.

Approaching the study of relations and their associated concepts demands a structured and methodical strategy. Begin by familiarizing yourself with the basic definitions and properties of relations, ensuring a solid grasp of the foundational concepts. Employing pictorial diagrams can significantly aid in visualizing complex relationships, making abstract ideas

more accessible. Practice is key; working through varied examples and problems will deepen understanding and reinforce learning. It is also beneficial to draw parallels between mathematical relations and their practical applications in computer science, such as database relations and function mappings. Engaging in group discussions and seeking clarifications on challenging concepts can provide new insights and foster a deeper understanding. Leveraging online resources, such as tutorials and educational videos, can also supplement learning by offering diverse perspectives and explanations. By adopting these strategies, students can build a robust understanding of relations and their applications, setting a strong foundation for future studies and professional endeavours in computer applications.

1.1 Learning Objectives

At the end of this unit, you will be able to:

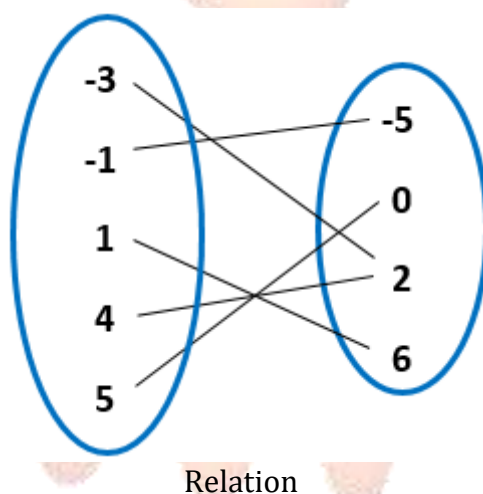
- ❖ *Recall the fundamental concept of a function, its components (domain and range), and the notation used to represent functions.*
- ❖ *Demonstrate an understanding of the relationships between inputs and outputs in a function.*
- ❖ *Determine the domain and range constraints for specific functions and critically evaluate scenarios where certain inputs might lead to undefined or complex outputs.*

2. RELATION, PICTORIAL DIAGRAMS, DOMAIN, CO-DOMAIN, RANGE OF A RELATION

2.1 Relations

A relation is a mathematical concept that describes how elements from one set are related to elements in another set. It defines connections or associations between these elements.

Based on the definition we can say that, a relation A is a subset of $A \times A$.



Example:

Consider a relation R in the set $A = \{1, 2, 3, 4\}$ given by $R = \{(a, b) : a - b = 10\}$. This is an empty set, as no pair (a, b) satisfies the condition $a - b = 10$.

Relations are used to represent interactions or connections between different objects or entities. In mathematics, a relation is typically expressed as a set of ordered pairs, where each pair consists of an element from the first set (called the domain) and an element from the second set (called the co-domain). Relations can be used to model a wide range of real-world scenarios, such as the relationships between people, numbers, or any other objects.

2.1.1 Types of Relations

1. Empty Relations

A relation R in a set A is called empty relation, if no element of A is related to any element of A , i.e., $R = \phi \subset A \times A$.

2. Universal Relation

A relation R in a set A is called universal relation, if each element of A is related to every element of A , i.e. $R = A \times A$

Both the empty universal Relation are also called trivial relations

Example:

Let A be the set of all students of a boys school. Show that the relation R in A given by $R = \{(a, b) : a \text{ is sister of } b\}$ is the empty relation and $R' = \{(a, b) : \text{the difference between heights of } a \text{ and } b \text{ is less than 3 meters}\}$ is the universal relation.

Solution:

Since the school is boys school, no student of the school can be sister of any student of the school. Hence, $R = \phi$, showing that R is the empty relation. It is also obvious that the difference between heights of any two students of the school has to be less than 3 meters. This shows that $R' = A \times A$ is the universal relation.

3. Reflexive, Symmetric, Transitive

A relation R in a set A is called

- (i) reflexive, if $(a, a) \in R$, for every $a \in A$,
- (ii) symmetric, if $(a_1, a_2) \in R$ implies that $(a_2, a_1) \in R$, for all $a_1, a_2 \in A$.
- (iii) transitive, if $(a_1, a_2) \in R$ and $(a_2, a_3) \in R$ implies that $(a_1, a_3) \in R$, for all $a_1, a_2, a_3 \in A$.

4. Equivalence Relation

A relation R in a set A is said to be an equivalence relation if R is reflexive, symmetric, and transitive.

Example:

Let T be the set of all triangles in a plane with R a relation in T given by $R = \{(T_1, T_2) : T_1 \text{ is congruent to } T_2\}$. Show that R is an equivalence relation. Solution R is reflexive since every triangle is congruent to itself. Further, $(T_1, T_2) \in R \Rightarrow T_1 \text{ is congruent to } T_2 \Rightarrow T_2 \text{ is congruent to } T_1 \Rightarrow (T_2, T_1) \in R$. Hence, R is symmetric. Moreover, $(T_1, T_2), (T_2, T_3) \in R \Rightarrow T_1 \text{ is congruent to } T_2 \text{ and } T_2 \text{ is congruent to } T_3 \Rightarrow T_1 \text{ is congruent to } T_3 \Rightarrow (T_1, T_3) \in R$. Therefore, R is an equivalence relation.

Example

Show that the relation R in the set Z of integers given by $R = \{(a, b) : 2 \text{ divides } a - b\}$ is an equivalence relation.

Solution:

R is reflexive, as 2 divides $(a - a)$ for all $a \in Z$. Further, if $(a, b) \in R$, then 2 divides $a - b$. Therefore, 2 divides $b - a$. Hence, $(b, a) \in R$, which shows that R is symmetric. Similarly, if $(a, b) \in R$ and $(b, c) \in R$, then $a - b$ and $b - c$ are divisible by 2. Now, $a - c = (a - b) + (b - c)$ is even (Why?). So, $(a - c)$ is divisible by 2. This shows that R is transitive. Thus, R is an equivalence relation in Z .

In the above example, note that all even integers are related to zero, as $(0, \pm 2)$, $(0, \pm 4)$ etc., lie in R and no odd integer is related to 0, as $(0, \pm 1)$, $(0, \pm 3)$ etc., do not lie in R . Similarly, all odd integers are related to one and no even integer is related to one. Therefore, the set E of all even integers and the set O of all odd integers are subsets of Z satisfying following conditions:

- (i) All elements of E are related to each other and all elements of O are related to each other.
- (ii) No element of E is related to any element of O and vice-versa.
- (iii) E and O are disjoint and $Z = E \cup O$.

The subset E is called the equivalence class containing zero and is denoted by $[0]$. Similarly, O is the equivalence class containing 1 and is denoted by $[1]$. Note that $[0] \neq [1]$, $[0] = [2r]$ and $[1] = [2r + 1]$, $r \in Z$. Infact, what we have seen above is true for an arbitrary equivalence relation R in a set X . Given an arbitrary equivalence relation R in an arbitrary set X , R divides X into mutually disjoint subsets A_i called partitions or subdivisions of X satisfying:

- (i) all elements of A_i are related to each other, for all i .
- (ii) no element of A_i is related to any element of A_j , $i \neq j$.
- (iii) $\cup A_j = X$ and $A_i \cap A_j = \phi$, $i \neq j$.

The subsets A_i are called equivalence classes. The interesting part of the situation is that we can go reverse also. For example, consider a subdivision of the set Z given by three mutually disjoint subsets A_1 , A_2 and A_3 whose union is Z with

$$A_1 = \{x \in \mathbb{Z} : x \text{ is a multiple of } 3\} = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

$$A_2 = \{x \in \mathbb{Z} : x - 1 \text{ is a multiple of } 3\} = \{\dots, -5, -2, 1, 4, 7, \dots\}$$

$$A_3 = \{x \in \mathbb{Z} : x - 2 \text{ is a multiple of } 3\} = \{\dots, -4, -1, 2, 5, 8, \dots\}$$

Define a relation R in $\mathbb{Z}$ given by $R = \{(a, b) : 3 \text{ divides } a - b\}$. Following the arguments similar to those used in Example 5, we can show that R is an equivalence relation. Also, A_1 coincides with the set of all integers in $\mathbb{Z}$ which are related to zero, A_2 coincides with the set of all integers which are related to 1 and A_3 coincides with the set of all integers in $\mathbb{Z}$ which are related to 2. Thus, $A_1 = [0]$, $A_2 = [1]$ and $A_3 = [2]$.

In fact, $A_1 = [3r]$, $A_2 = [3r + 1]$ and $A_3 = [3r + 2]$, for all $r \in \mathbb{Z}$.

2.2 Pictorial Diagrams

Pictorial diagrams are visual representations used to illustrate relations or connections between elements from different sets. They provide a graphical way to understand and present relationships.

Pictorial diagrams are valuable tools in mathematics and other fields to help people visualize complex relationships. In the context of relations, these diagrams can be used to represent the connections between elements in the domain and the co-domain. For example, you can use arrows, lines, or other symbols to show how elements from one set correspond to elements in another. Pictorial diagrams make abstract concepts more accessible and aid in problem-solving and comprehension.

2.3 Domain

The domain of a relation is the set of all input or independent values (elements) for which the relation is defined. It represents the possible input values that can be used to determine the relation's output values.

In a relation, the domain specifies the "starting point" or the values that you can plug into the relation to obtain corresponding output values. For example, in a relation that defines the age of individuals as a function of their names, the domain might be the set of names of all individuals. The domain is essential for understanding the scope of a relation and for ensuring that you use valid inputs.

2.4 Co-Domain

The co-domain of a relation is the set of all possible output or dependent values (elements) that the relation can produce. It represents the full range of potential output values, regardless of whether they are actually produced by the relation.

The co-domain defines the "destination" or the set of possible output values that the relation can map elements from the domain to. It is important to note that not all elements from the co-domain may have a corresponding element in the range (the actual output values). The co-domain is a broader concept, while the range describes the actual output values produced by the relation. Understanding the co-domain helps define the potential outcomes of a relation.

2.5 Range of a Relation

The range of a relation is the set of all actual output values (elements) produced by the relation when elements from the domain are used as input. It represents the subset of the co-domain that is "covered" or "occupied" by the relation.

The range is a subset of the co-domain and defines the actual output values generated by the relation for the given inputs. It is essential for understanding the specific values produced by the relation and its behaviour. The range may be smaller than the co-domain if not all possible outputs are achieved. It is a crucial aspect of the relation that characterizes its behaviour and the set of values it affects.

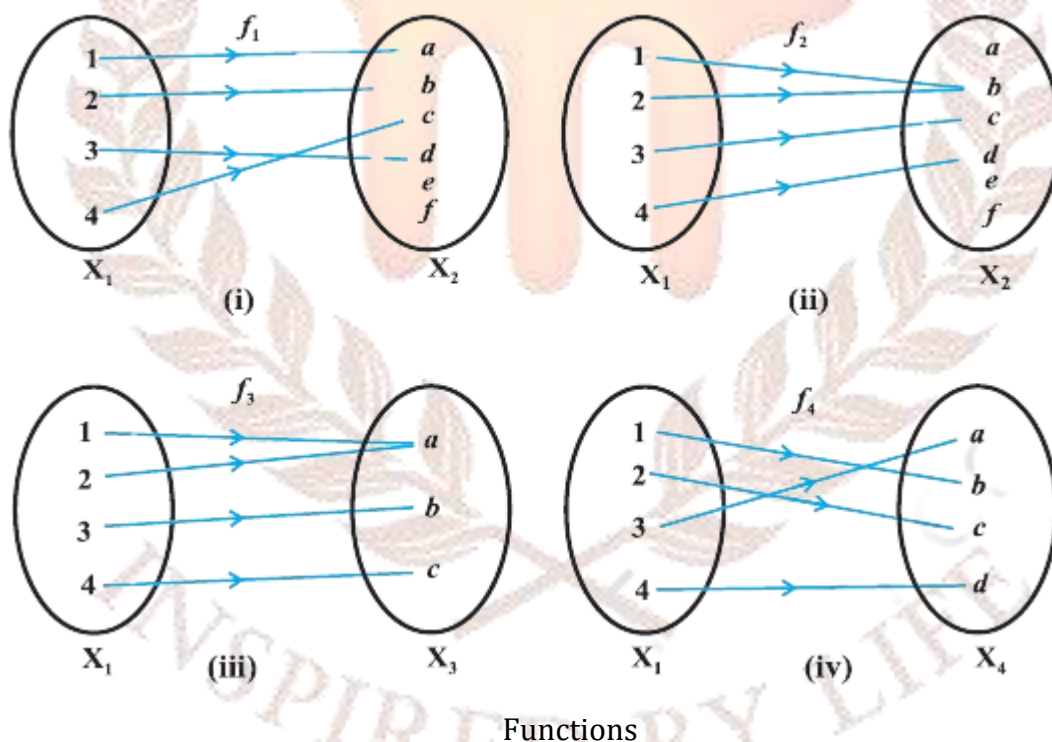
3. FUNCTION

A function is a mathematical relationship that assigns each input element from a set (called the domain) to a unique output element in another set (called the codomain). It can be represented by a rule or equation and is denoted as,

$$f(x) \text{ or } y = f(x)$$

Functions provide a structured way to understand how variables change with respect to each other.

Whether we're analysing population growth, predicting stock market trends, or modelling the trajectory of a projectile, functions provide us with a clear framework to decipher and manipulate complex relationships.



Definition 1: A function $f : X \rightarrow Y$ is defined to be one-one (or injective), if the images of distinct elements of X under f are distinct, i.e., for every $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. Otherwise, f is called many-one.

The function f_1 and f_4 in above figure (i) and (iv) are one-one and the function f_2 and f_3 in the figure (ii) and (iii) are many-one.

Definition 2: A function $f : X \rightarrow Y$ is said to be onto (or surjective), if every element of Y is the image of some element of X under f , i.e., for every $y \in Y$, there exists an element x in X such that $f(x) = y$.

The function f_3 and f_4 in the above figure (iii), (iv) are onto and the function f_1 in the figure (i) is not onto as elements e, f in X_2 are not the image of any element in X_1 under f_1 .

Remark $f : X \rightarrow Y$ is onto if and only if $\text{Range of } f = Y$.

Definition 3: A function $f : X \rightarrow Y$ is said to be one-one and onto (or bijective), if f is both one-one and onto.

The function f_4 in the figure (iv) is one-one and onto.

Example:

Let N be the set of natural numbers and the relation R be defined on N such that $R = \{(x, y) : y = 2x, x, y \in N\}$.

What is the domain, codomain and range of R ? Is this relation a function?

Solution:

The domain of R is the set of natural numbers N . The codomain is also N . The range is the set of even natural numbers. Since every natural number n has one and only one image, this relation is a function.

4. DOMAIN AND RANGE OF A FUNCTION

Domain of a Function:

The domain of a function is the set of all possible input values for which the function is defined. It's the collection of values that you can plug into the function to get meaningful output. For example, in the function

$$f(x) = \sqrt{x}$$

the domain would include all non-negative real numbers, as the square root of a negative number is undefined in the real number system.

Range of a Function:

The range of a function is the set of all possible output values that the function can produce based on its domain. It represents the values that the function "maps" its domain elements to. In the function,

$$f(x) = x^2$$

the range consists of all non-negative real numbers, as the function squares the input values, resulting in non-negative outputs.

Consider a function that serves as a translator between the languages of inputs and outputs. The collection of legitimate inputs for which the function is defined is known as the domain; these are the inputs that make sense under the circumstances of the issue. The range, meanwhile, covers all potential outputs that the function may produce in accordance with its domain. Functions "translate" inputs into equivalent outputs, enabling us to understand how changes in one quantity influence another.

4.1 Finding Domain and Range of a Linear Function

Consider the function

$$f(x) = 2x - 3$$

Domain: The domain of a function usually includes all the real numbers unless there are specific restrictions. In this case, there are no restrictions mentioned, so the domain is all real numbers: $(-\infty, \infty)$.

Range: To find the range, we can consider the fact that this is a linear function. As x varies from negative infinity to positive infinity, the output $(f(x))$ will also vary across all real numbers. Therefore, the range is also all real numbers: $(-\infty, \infty)$.

Finding Domain and Range of a Square Root Function

Consider the function

$$g(x) = \sqrt{4 - x^2}$$

Domain: For a square root function, the expression inside the square root must be non-negative to avoid taking the square root of a negative number. So, solve the inequality: $4 - x^2 \geq 0$. This leads to $-2 \leq x \leq 2$. Thus, the domain is $[-2, 2]$.

Range: Since the function involves taking the square root of a non-negative value, the range will be all non-negative real numbers: $[0, \infty)$.

Finding Domain and Range of an Absolute Value Function

Consider the function

$$h(x) = |2x - 5|.$$

Domain: Absolute value functions have no restrictions on their domain, so the domain is all real numbers: $(-\infty, \infty)$.

Range: The absolute value function results in non-negative values. As $2x - 5$ varies from negative infinity to positive infinity, the absolute value will always be non-negative. So, the range is $[0, \infty)$.

Finding Domain and Range of a Quadratic Function

Consider the function

$$p(x) = x^2 - 4x + 3.$$

Domain: There are no restrictions on the domain of a quadratic function, so the domain is all real numbers: $(-\infty, \infty)$.

Range: To find the range of a quadratic function, consider the vertex. The vertex is at $x = 2$ (found by completing the square). As x^2 is always non-negative, the lowest value the function can have is 0. Therefore, the range is $[0, \infty)$.

4.2 Types of Functions

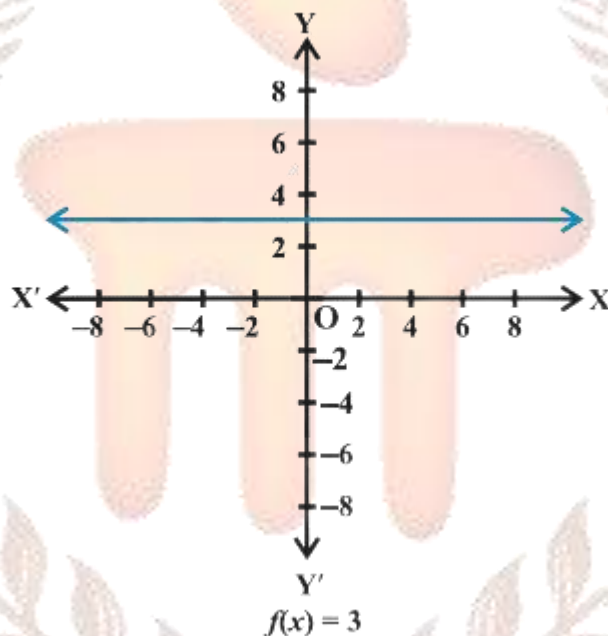
4.2.1 Real-Valued Functions

A real-valued function is a function that assigns real numbers as output values for real numbers in its domain. In other words, it takes real numbers as input and produces real numbers as output.

Real-valued functions are widely used in mathematics, science, and engineering to model various phenomena. They can represent relationships, measurements, and behaviours in the real world, making them essential for applications like physics, economics, and more.

4.2.2 Constant Function

Define the function $f: \mathbb{R} \rightarrow \mathbb{R}$ by $y = f(x) = c$, $x \in \mathbb{R}$ where c is a constant and each $x \in \mathbb{R}$. Here domain of f is $\mathbb{R}$ and its range is $\{c\}$.



Constant Function

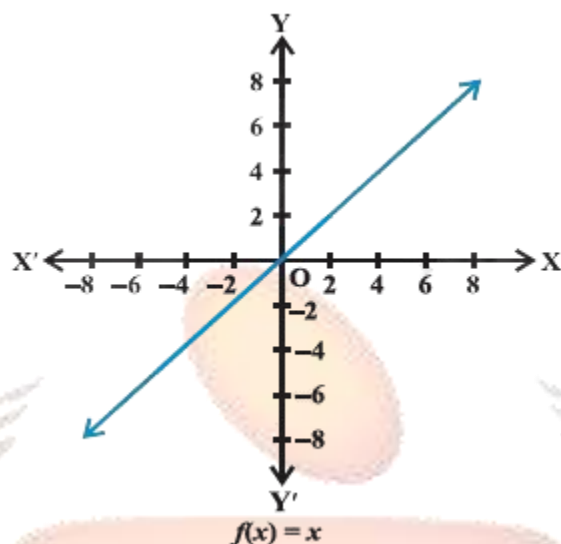
Domain: The domain of a constant function is the set of all real numbers.

Range: The range of a constant function is a single real number, which is the constant value itself.

Example: $f(x)=3$ has a domain of all real numbers and a range of $\{3\}$.

4.2.3 Identity Function

Let $\mathbb{R}$ be the set of real numbers. Define the real valued function $f: \mathbb{R} \rightarrow \mathbb{R}$ by $y = f(x) = x$ for each $x \in \mathbb{R}$. Such a function is called the identity function. Here the domain and range of f are $\mathbb{R}$. The graph is a straight line as shown in below. It passes through the origin.



Identity Function

Domain: The domain of the identity function is the set of all real numbers.

Range: The range of the identity function is also the set of all real numbers.

Example: $f(x)=x$ has a domain and range of all real numbers.

4.2.4 Polynomial Function:

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be polynomial function if for each x in $\mathbb{R}$, $y = f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where n is a non-negative integer and $a_0, a_1, a_2, \dots, a_n \in \mathbb{R}$.

The functions defined by $f(x) = x^3 - x^2 + 2$, and $g(x) = x^4 + \sqrt{2}x$ are some examples of polynomial functions, whereas the function h defined by $h(x) = x^{3/2} + 2x$ is not a polynomial function.

Domain: The domain of a polynomial function is the set of all real numbers.

Range: The range of a polynomial function can vary depending on the degree and leading coefficients of the polynomial. It can cover a wide range of real numbers.

Example: $f(x)=2x^3-5x^2+4x-1$ has a domain and range of all real numbers.

4.2.5 Rational Function

Rational functions are functions of the type $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomial functions of x defined in a domain, where $g(x) \neq 0$.

Domain: The domain of a rational function includes all real numbers except the values that make the denominator equal to zero, as division by zero is undefined.

Range: The range of a rational function can also vary, typically covering a wide range of real numbers.

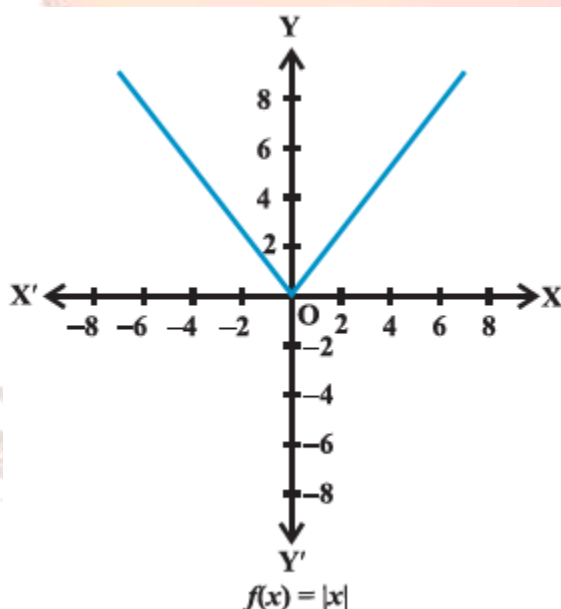
Example: $f(x)=1/x$ has a domain of all real numbers except $x=0$, and its range includes all real numbers except zero.

4.2.6 Modulus Function (Absolute Value)

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ for each $x \in \mathbb{R}$ is called modulus function. For each non-negative value of x , $f(x)$ is equal to x . But for negative values of x , the value of $f(x)$ is the negative of the value of x , i.e.,

$$f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Modulus Function



Domain: The domain of the modulus function is all real numbers.

Range: The range of the modulus function is the set of non-negative real numbers (values greater than or equal to zero).

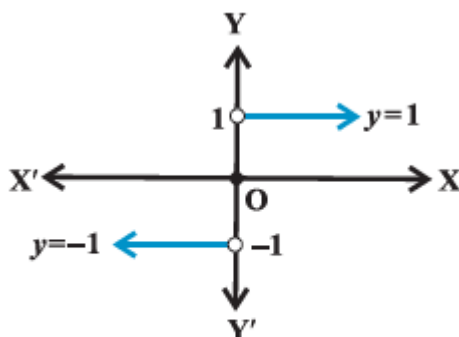
Example: $f(x)=|x|$ has a domain of all real numbers and a range of non-negative real numbers.

4.2.7 Signum Function (Sign Function)

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

is called the signum function. The domain of the signum function is $\mathbb{R}$ and the range is the set $\{-1, 0, 1\}$.



$$f(x) = \frac{|x|}{x}, x \neq 0 \text{ and } 0 \text{ for } x = 0$$

Signum Function

Domain: The domain of the signum function is all real numbers.

Range: The range of the signum function consists of three values: -1, 0, and 1. It represents the sign or direction of a real number.

Example: $f(x) = \text{sgn}(x)$ has a domain of all real numbers and a range of $\{-1, 0, 1\}$.

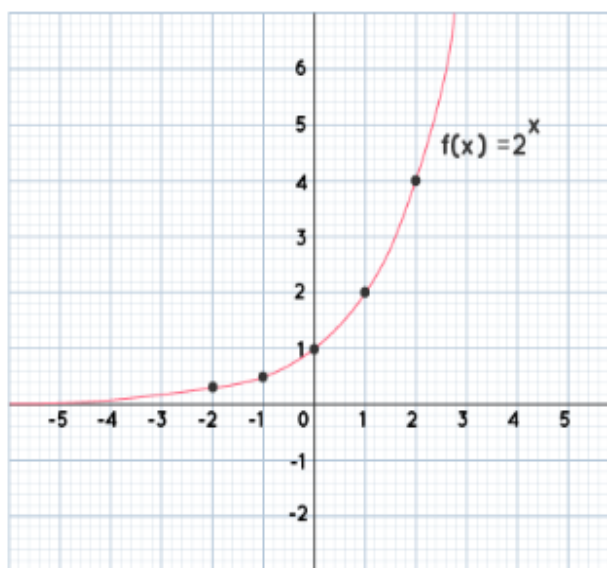
4.2.8 Exponential Function

Exponential function, as its name suggests, involves exponents. But note that, an exponential function has a constant as its base and a variable as its exponent.

$f(x) = 2^x$

x	f(x)
-2	0.25
-1	0.5
0	1
1	2
2	4

$2^{-2} = \frac{1}{4} = 0.25$
 $2^{-1} = \frac{1}{2} = 0.5$
 $2^0 = 1$
 $2^1 = 2$
 $2^2 = 4$



Exponential Function

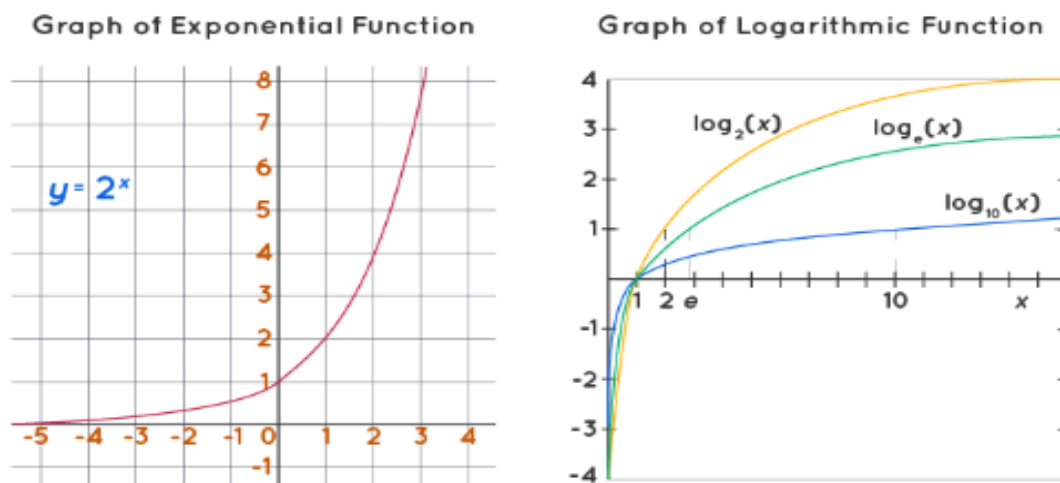
Domain: The domain of an exponential function is all real numbers.

Range: The range of an exponential function depends on its base and whether it is exponential growth or decay. It typically includes all positive real numbers.

Example: $f(x)=2^x$ has a domain and range of all real numbers, but the range is limited to positive values for $x>0$.

4.2.9 Logarithmic Function

The basic logarithmic function is of the form $f(x) = \log_a x$ (r) $y = \log_a x$, where $a > 0$.



Logarithmic Function

Domain: The domain of a logarithmic function is the set of all positive real numbers, as the logarithm of zero or a negative number is undefined.

Range: The range of a logarithmic function is all real numbers.

Example: $f(x)=\log(x)$ has a domain of all positive real numbers and a range of all real numbers.

4.2.10 Greatest Integer Function (Step Function)

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = [x]$, $x \in \mathbb{R}$ assumes the value of the greatest integer, less than or equal to x . Such a function is called the greatest integer function.

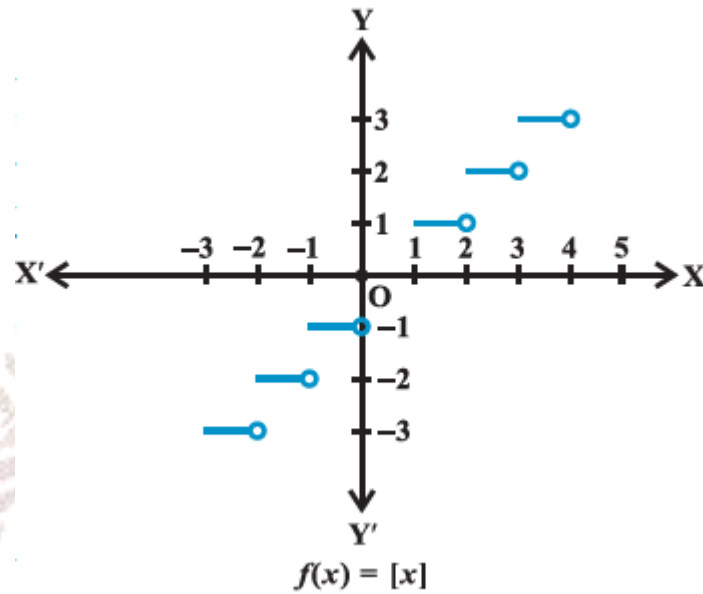
From the definition of $[x]$, we can see that

$$[x] = -1 \text{ for } -1 \leq x < 0$$

$$[x] = 0 \text{ for } 0 \leq x < 1$$

$$[x] = 1 \text{ for } 1 \leq x < 2$$

$[x] = 2$ for $2 \leq x < 3$ and so on...



Greatest Integer Function

Domain: The domain of the greatest integer function is all real numbers.

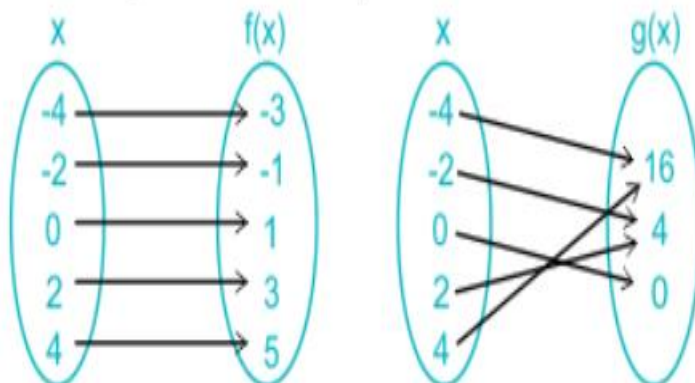
Range: The range of the greatest integer function consists of all integers, as it rounds down to the nearest integer for each real number.

Example: $f(x)=[x]$ has a domain of all real numbers and a range of all integers.

5. PROPERTIES OF FUNCTIONS

One-to-One Function (Injective):

Definition: A function is one-to-one if, for every distinct pair of elements a and b in the domain, the function assigns distinct values to them, i.e., $f(a) \neq f(b)$.

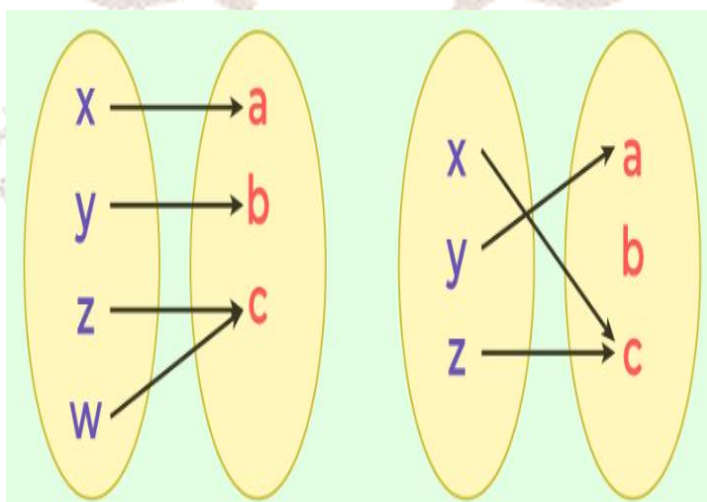


One-to-One Function

Example: The function $f(x)=2x$ is one-to-one because it maps distinct inputs to distinct outputs.

Onto Function (Surjective):

Definition: A function is onto if, for every element in the co-domain, there is at least one element in the domain that maps to it. In other words, the function covers the entire co-domain.



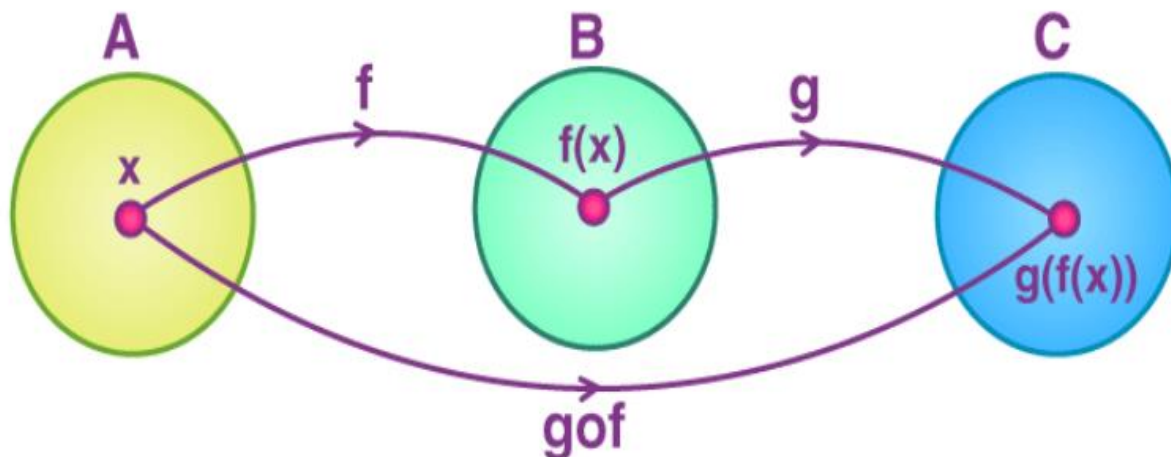
Onto Function

Example: The function $f(x)=x^2$ from the set of real numbers to the set of non-negative real numbers is onto.

Composite Function:

Definition: A composite function is formed by applying one function to the output of another function. If f and g are functions, then the composite function $f \circ g$ is defined as

$$(f \circ g)(x) = f(g(x)).$$

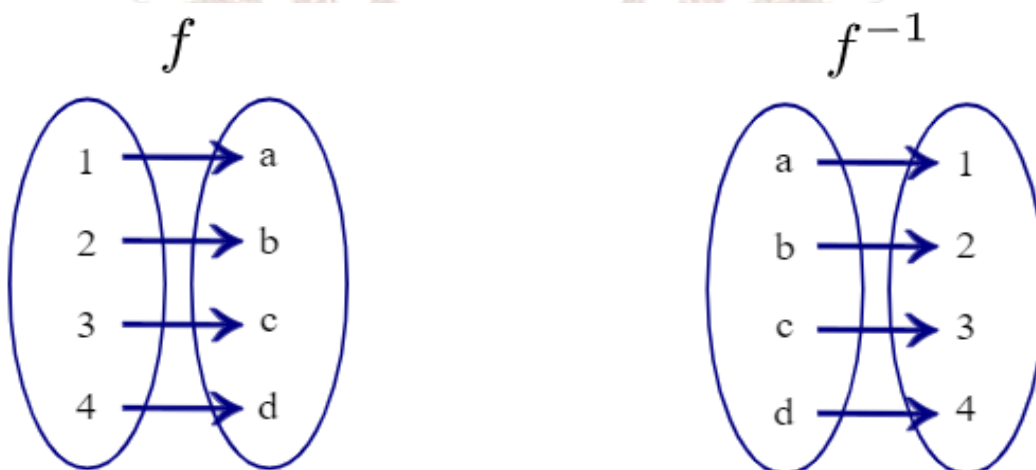


Composite Function

Example: If $f(x) = 2x$ and $g(x) = x + 3$, then $(f \circ g)(x) = 2(x + 3) = 2x + 6$.

Inverse of a Function:

Definition: The inverse of a function f is another function denoted as f^{-1} such that $f(f^{-1}(x)) = x$ for all x in the domain of f , and $f^{-1}(f(x)) = x$ for all x in the domain of f^{-1} .



Inverse Function

Example: The inverse of the function $f(x) = 3x$ is $f^{-1}(x) = x/3$, as

$$f(f^{-1}(x)) = 3(x/3) = x.$$

Binary Operation:

Definition: A binary operation is a function that takes two elements from a set and combines them to produce another element in the same set. It associates two elements with a third.

Example: Addition (+) and multiplication ($\times$) are binary operations on the set of real numbers because adding or multiplying two real numbers results in another real number.

6. SUMMARY

In our exploration of the fundamental mathematical concepts crucial for a BCA curriculum, we delve into the intriguing world of relations and their multifaceted applications in computer science. Let's unwrap this together, shall we?

At the heart of it, relations serve as a mathematical lens through which we can view and analyze the connections between different sets of data. Imagine you're organizing a library. Each book can be related to another based on genre, author, or publication year. In mathematics, this is akin to understanding how elements from one set relate to elements in another, painting a picture of interconnectedness that's vital in computing for organizing databases or developing efficient algorithms.

Now, let's get a bit more visual with pictorial diagrams. These are not just fancy drawings; they're powerful tools that help us visualize complex relationships in a simple and understandable way. Think of them as maps that guide us through the terrain of data relationships, making the journey from concept to application smoother.

Diving deeper, we encounter domains and co-domains, concepts that define the "where" of our relations – the starting and potential ending points. In a way, they set the stage for the actors (elements) to play out the drama of relations. The domain is all about possibilities, the different inputs we can use, while the co-domain is about potentialities, the myriad outcomes that might emerge.

The range, on the other hand, shows us the reality – the actual outcomes we get from our inputs. It's like after throwing a party, the range would be the guests who actually showed up, giving us a snapshot of the real connections formed.

Unit 1 of our journey isn't just academic; it's a toolbox that computer application students can use to craft solutions, model data, and essentially, think in a structured, logical manner. It's about seeing beyond numbers and sets, into the real-world applications like database structures and algorithm design.

So, how does one best navigate this landscape? Picture yourself as a mathematical explorer. Start with the basics, understand the terrain (definitions and properties), use your map (pictorial diagrams), and practice, practice, practice. Each problem solved is a territory charted. And remember, in this exploration, collaboration is key. Discussing with peers, seeking clarification, and utilizing diverse resources can unveil new pathways and insights.

7. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. What is the purpose of a function in mathematics?
 - a) To create complex relationships between different sets of variables.
 - b) To assign each input element to multiple output elements.
 - c) To establish a one-to-one correspondence between two sets.
 - d) To provide a structured way to understand how variables change in relation to each other.
2. How is a function typically denoted in mathematical notation?
 - a) $x = f(y)$
 - b) $y = f(x)$
 - c) $x + y = f$
 - d) $f(x) = y$
3. What does the domain of a function represent?
 - a) The set of all possible output values that the function can produce.
 - b) The set of all possible input values for which the function is defined.
 - c) The set of all possible input and output pairs.
 - d) The set of all non-negative real numbers.

4. In the function $f(x) = \sqrt{x}$, what values are included in the domain?
 - a) All real numbers, including negative and positive values.
 - b) All non-negative real numbers.
 - c) All positive real numbers.
 - d) All integer values.
5. What does the range of a function represent?
 - a) The set of all possible input values for which the function is defined.
 - b) The set of all possible output values that the function can produce.
 - c) The set of all non-negative real numbers.
 - d) The set of all positive real numbers.
6. How is the concept of domain and range related to functions being translators between inputs and outputs?
 - a) The domain represents the set of all possible translations, and the range represents the translated outputs.
 - b) The domain and range both represent the same set of values.
 - c) The domain represents the set of legitimate inputs, and the range represents the set of all possible inputs and outputs.
 - d) The domain represents the set of legitimate inputs, and the range represents the set of potential outputs that correspond to those inputs.
7. What is a defining characteristic of an injective function?
 - a) Every element in the domain has multiple pre-images in the codomain.
 - b) Every element in the codomain has at least one pre-image in the domain.
 - c) Each distinct element in the domain maps to a distinct element in the codomain.
 - d) Multiple elements from the domain map to a single element in the codomain.

8. How can the concept of an injective function be visualized?
 - a) By representing elements from the domain and codomain using symbols.
 - b) By arranging elements in the codomain in a specific order.
 - c) By drawing arrows between the domain and the codomain on a graph.
 - d) By imagining a conveyor belt where items from the domain are uniquely assigned to items in the codomain.
9. What is a defining characteristic of an "onto" function?
 - a) Every element in the domain has exactly one pre-image in the codomain.
 - b) Every element in the codomain has exactly one pre-image in the domain.
 - c) Multiple elements from the domain map to a single element in the codomain.
 - d) Every element in the codomain has at least one pre-image in the domain.
10. How can the concept of "onto" functions be visualized?
 - a) By arranging elements in the codomain in a specific order.
 - b) By imagining a shooting range with targets for each domain element.
 - c) By drawing arrows between the domain and the codomain on a graph.
 - d) By representing elements from the domain and codomain using symbols.
11. What does an "into" function ensure?
 - a) Each element in the codomain has exactly one pre-image in the domain.
 - b) Each element in the domain maps to a unique element in the codomain.
 - c) Multiple elements from the domain map to a single element in the codomain.
 - d) Elements from the domain map to elements in the codomain without any restrictions.

12. How does an "into" function distribute elements from the domain to the codomain?

- a) Each element from the domain maps to multiple elements in the codomain.
- b) Each element from the domain maps to exactly one element in the codomain.
- c) Multiple elements from the codomain map to a single element in the domain.
- d) The distribution is random and doesn't follow any pattern.

8. TERMINAL QUESTIONS

1. Define a function in mathematics.
2. What does the domain of a function represent?
3. Explain the concept of the range of a function.
4. How is a function typically denoted in mathematical notation?
5. Provide an example of a function and its domain and range.
6. Explain the difference between the domain and the range of a function.
7. What does it mean for a function to be "one-to-one" or "injective"?
8. How is an "onto" function different from an "into" function?
9. Give an example of a real-world situation that can be modelled using an "onto" function.
10. Define an "into" function and provide an example.
11. Describe the purpose of a function in mathematics and how it helps us understand relationships between variables.
12. Explain the process of finding the domain of a function. Provide an example.
13. Describe how to determine the range of a function. Use an example to illustrate.
14. Compare and contrast the concepts of "one-to-one" (injective), "onto" (surjective), and "into" functions.
15. Consider a function that maps students to their grades in a class. Discuss how the domain and range would be defined for this function.
16. Choose a real-world scenario and describe how you can represent it using a function. Include details about the domain, codomain, and mapping.
17. Discuss the significance of an "into" function and provide a real-world example that fits this description.

18. Imagine a function that models the height of a growing tree over time. Explain how the domain and range would be determined for this function.
19. Explain the visual analogy used to describe an "onto" function. Provide another analogy to help visualize the concept of a function.
20. Take a quadratic function and analyse its domain and range using both algebraic and graphical methods

9. ANSWERS

Self-Assessment Questions

1. Answer: d) To provide a structured way to understand how variables change in relation to each other.
2. Answer: b) $y = f(x)$
3. Answer: b) The set of all possible input values for which the function is defined.
4. Answer: b) All non-negative real numbers.
5. Answer: b) The set of all possible output values that the function can produce.
6. Answer: d) The domain represents the set of legitimate inputs, and the range represents the set of potential outputs that correspond to those inputs.
7. Answer: c) Each distinct element in the domain maps to a distinct element in the codomain.
8. Answer: c) By drawing arrows between the domain and the codomain on a graph.
9. Answer: d) Every element in the codomain has at least one pre-image in the domain.
10. Answer: c) By drawing arrows between the domain and the codomain on a graph.
11. Answer: d) Elements from the domain map to elements in the codomain without any restrictions.
12. Answer: b) Each element from the domain maps to exactly one element in the codomain.

Terminal Questions

1. Refer to the section titled "Function" where a function is introduced as a mathematical relationship that assigns each input element from a set to a unique output element in another set.
2. Look under the subheading "Domain of a Function" which explains the domain as the set of all possible input values for which the function is defined.

3. The "Range of a Function" section will provide insights into the range, described as the set of all possible output values the function can produce based on its domain.
4. In the initial paragraphs under "Function," functions are denoted using notations like $f(x)$ or $y = f(x)$, illustrating how functions are typically expressed in mathematics.
5. The "Example" sections under both "Domain and Range of a Function" offer clear examples of functions along with their respective domains and ranges.
6. This difference is discussed in the "Domain of a Function" and "Range of a Function" sections, highlighting how the domain is about input values while the range focuses on output values.
7. The "One-to-One Function (Injective)" section under "Properties of Functions" delves into the definition and characteristics of injective functions.
8. Differences between "onto" and "into" functions can be deduced from the descriptions in the "Onto Function (Surjective)" and the context provided for "into" functions across the content.
9. Infer from the "Onto Function (Surjective)" section how any situation where each output is guaranteed to have an input can be modelled as an "onto" function.
10. "Into" functions are implicitly described in contrast to "onto" functions within the content. An "into" function might not map every element in the codomain to an element in the domain, which can be inferred from the broader discussion on functions.
11. The introductory paragraphs of the "Function" section shed light on this, explaining how functions provide a structured way to understand variable interrelationships.
12. Under "Domain of a Function," the process is outlined, and specific examples are given to illustrate how to determine a function's domain.
13. The "Range of a Function" section discusses how to identify the range, complemented by examples that clearly demonstrate the concept.
14. These concepts are elaborated upon in sections like "One-to-One Function (Injective)," and "Onto Function (Surjective)," while "into" functions are implied through the explanations of these properties. Combining information from these sections will help compare and contrast the concepts.
15. Applying the principles from "Domain of a Function" and "Range of a Function," one can deduce how the domain and range would apply to this scenario, even though it's not explicitly described in the content.

16. While not explicitly outlined, combining insights from "Function," "Domain and Range of a Function," and examples provided can help construct a representation for a real-world scenario.
 17. The significance can be inferred from the broader discussions on functions, particularly by understanding what is elaborated under "onto" functions and reversing the concept to understand "into" functions.
 18. By synthesizing concepts from "Domain and Range of a Function," one can apply these to model the growth of a tree, using time as the domain and height as the range.
 19. The analogy for an "onto" function can be drawn from the explanations under "Onto Function (Surjective)," where each element in the codomain is mapped from the domain, similar to hitting every target in a shooting range.
 20. The principles under "Domain and Range of a Function" can be applied, alongside the mathematical and graphical interpretations discussed throughout the function sections.
- 